CONTENTS

ABSTRACT

Object detection based on deep learning is a popular trend, and it includes object recognition and positioning. This paper proposes a method that can accurately obtain object type and accurate threedimensional position. The method is divided into three parts: object recognition and coarse positioning based on deep learning, precise positioning based on deep learning combined with B-spline level set in color images, and precise three-dimensional positioning with depth information of RGB-D camera. The precise positioning of the object provides accurate end pose information for the autonomous grasping of the robotic arm, and it has great significance to the gripping of the robotic arm. Performance metrics include MAP (mean average precision) and IOU (intersection of a union). Experimental results show that the MAP value of Yolo-v3 in this paper can reach 87.62%, the average IOU of Yolo-v3 in this paper can reach 66.74%, the average IOU of Yolo-v3 and B-spline level set can reach 100%, and can get accurate 3D location in the real scene. In addition, the experiments comparisons between VOC dataset and our own dataset validate that our dataset can take higher MAP and average IOU values.

INTRODUCTION

Object detection, a fundamental task in computer vision, has evolved from traditional region-based methods using handcrafted features to deep learning-based approaches leveraging convolutional neural networks (CNNs). Unlike traditional methods that rely on manually extracted features, deep learning models learn and extract high-dimensional features directly from images, improving robustness and accuracy. Object detection not only classifies objects but also determines their precise locations, making it essential for applications like autonomous driving, medical imaging, and robotics. Modern approaches are categorized into two-stage methods (e.g., R-CNN, Faster R-CNN) and single-stage methods (e.g., YOLO, SSD), with the latter offering improved real-time performance. Recent advancements integrate RGB-D cameras and level-set methods to enhance 3D positioning accuracy, further optimizing applications in robotics and intelligent systems.

Deep learning has revolutionized object detection by enabling automated feature extraction and faster, more accurate recognition. While traditional methods like HOG, SIFT, and SVM were computationally intensive and less adaptable, modern CNN-based models such as YOLO and SSD offer real-time detection with superior precision. Additionally, integrating RGB-D cameras and advanced algorithms like the level set method enhances 3D object positioning, making it crucial for applications in robotics, surveillance, and autonomous navigation. The continuous evolution of object detection techniques aims to improve efficiency, reduce computational costs, and expand their applicability in real-world scenarios.

LITERATURE SURVEY

Paper and Author details	Description	Advantages	Disadvantages
Authors: A. Ilioudi, A. Dabiri, B. J. Wolf, and B. De Schutter **Title:** "Deep learning for object detection and segmentation in videos: Toward an integration with domain knowledge," **Year:** 2024	The study by Ilioudi et al. (2024) explores deep learning-based object detection and segmentation in videos, emphasizing integration with domain knowledge for improved accuracy and adaptability.	Enhanced precision in detecting and segmenting objects, improved performance through domain-specific insights, and better real-time adaptability for dynamic environments.	Computational requirements, dependency on large annotated datasets, and challenges in generalizing across diverse video contexts.
Authors: L. Jiao, F. Zhang, F. Liu, S. Yang, L. Li, Z. Feng, and R. Qu, **Title:** ''A survey of deep learning-based object detection,'' **Year:** 2023	Jiao et al. (2023) provide a comprehensive survey on deep learning-based object detection, highlighting advancements, challenges, and future directions. The study discusses how CNN-based models like Faster R-CNN, YOLO, and SSD outperform traditional methods by automating feature extraction and improving detection accuracy.	high precision, adaptability to complex environments, and real-time processing capabilities.	Deep learning models require large datasets, significant computational resources, and may struggle with small object detection and occlusions
Authors: L. Kalake, W. Wan, and L. Hou, **Title:** "Analysis based on recent deep learning approaches applied in real-time multi-object tracking" **Year:** 2023	The study by L. Kalake, W. Wan, and L. Hou (2023) reviews recent deep learning approaches for real-time multi-object tracking, analyzing their effectiveness and challenges. These methods leverage CNNs and transformer-based architectures to enhance tracking accuracy and robustness in dynamic environments.	Improved feature extraction, real-time performance, and adaptability to occlusions and complex backgrounds	High computational demands, sensitivity to lighting variations, and potential failure in handling fast-moving or overlapping objects.

Authors: H. Sun, W. Zhang, R. Yu, and Y. Zhang, **Title:** "Motion planning for mobile robots-focusing on deep reinforcement learning: A systematic review," **Year:** 2022	The paper by Sun et al. (2022) provides a systematic review of motion planning for mobile robots, emphasizing deep reinforcement learning (DRL). It explores various DRL-based approaches that enhance robot navigation by enabling adaptive decision-making in dynamic environments	DRL in motion planning include improved adaptability, real-time learning from interactions, and the ability to handle complex scenarios without explicit programming.	High computational demands, long training times, and challenges in ensuring safety and reliability in unpredictable real-world settings.

PROBLEM STATEMENT

The traditional object detection approach relies on hand-crafted features and region-based classification, which suffer from complexity and poor robustness. Deep learning, particularly CNN-based methods, has significantly improved object detection by learning features directly from raw images. However, precise object positioning remains a challenge. This study enhances object detection and positioning by integrating the level set method for color images with YOLO-v3 feedback and RGB-D depth data, enabling accurate 3D localization for applications like robotic arm grasping.

OBJECTIVES

The primary objectives of this seminar are:

- **Review of Object Detection Methods**
    - Analyze traditional object detection algorithms, including sliding window approaches and hand-crafted feature extraction.
    - Compare these methods with deep learning-based object detection techniques.
- **Investigation of Deep Learning in Object Detection**
    - Explore the advantages of convolutional neural networks (CNNs) in object detection.
    - Study how deep learning improves feature extraction, classification, and positioning of objects.
- **Comparison of Two-Stage and Single-Stage Object Detection Models**
    - Discuss the characteristics, strengths, and limitations of two-stage models (R-CNN, Fast R-CNN, Faster R-CNN).
    - Analyze single-stage models (SSD, YOLO) and their impact on real-time object detection.
- **Object Recognition and Positioning Enhancement**
    - Evaluate the use of RGB-D cameras for accurate object positioning.
    - Improve object positioning by integrating deep learning models with depth information.
- **Improvement of Object Detection and Positioning Using Level Set Method**
    - Modify the traditional level set algorithm to be applicable to color images.
    - Use level set outputs as feedback for YOLO-v3 to refine object positioning.
- **Development of a 3D Object Positioning System**
    - Combine RGB information with depth data to obtain precise 3D object locations.
    - Provide reliable position information for applications like robotic arm grasping.
- **Performance Evaluation and Analysis**
    - Assess detection accuracy, real-time performance, and computational efficiency of the proposed approach.
    - Compare results with existing object detection and positioning techniques.
- **Future Research Directions**
    - Explore ways to reduce hardware constraints for YOLO-v3 and similar models.
    - Investigate further improvements in object detection and positioning using advanced deep learning techniques.

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